

* What is Sorting?

Arranging data in specific order based on some parameter.

Eg.

3 8 9 14 19 - Ascending

By value

1 3 5 9 6 10 12 - Ascending by

1 - 2 2 3 4 4 6 no. of factors

* Min Cost to remove all elements:

0 1 2

Ex. $a[3] = 2 \quad 1 \quad 4$

remove 1: $[2 \ 1 \ 4] = 2 + 1 + 4 = 7$

remove 2: $[2 \ 4] = 2 + 4 = 6$

remove 4: $[4] = 4$
 17

Observation:

we have to delete the elements in descending order of

value to get min cost.

[a b c d]

$$\text{min cost} = a + 2b + 3c + 4d$$

Tc: $O(N \log N)$

Sc: $O(1)$

* Noble Integer:-

(No. of elements $< A[i]$) = $A[i]$

Example-

arr[] = 1 5 3 5 -10 2

count of $\Rightarrow$ 2 1 3 5 0 4

elements $< A[i]$

ans = 3

Tc : $O(N \log N)$

Sc : $O(1)$

* Noble Integer:-

Note :- data can repeat

(No. of elements $< A[i]$) = $A[i]$

arr[] = -10 1 1 3 100

count(A[i]): 0 1 1 3 4

un = 3

★ if element is same as previous
count will be not change.

if $A[i] == A[i-1]$

count won't change

A =	-10	1	1	2	2	4	4	8	10
Count:	0	1	1	3	4	4	4	7	8
Index	0	1	2	3	4	5	6	7	8

When an element comes for first time
no. of elements less than $A[i] = \text{index}$

if $(A[i] \neq A[i-1])$

Count = i

Tc: $O(N \log N)$

Sc: $O(1)$

* Comparators.:

Example

	9	3	4	8	16	37	6	13	15
factors	3	2	3	4	5	2	4	2	4

Order = 3 13 37 4 9 6 8 15 16

At every step, sorting algo will pick 2 elements & it will compare them. Based on the comparison, it will decide whether to rearrange them or not.

Above process will repeat till the data is sorted.

if you want a to come first

→ return -ve

if you want b to come first

→ return +ve

Tc: $O(N \log N)$

Sc: $O(N)$